

AAC Time Bridge

NodeMCU ESP-32S / ESP-WROOM-32 pin-assignment reference

REV 1 | 26 SEP 2026

US LETTER - LANDSCAPE | 1 / 1

Board / physical-pin convention: 38-pin board, two 19-pin rows. Top/component view, antenna at top and USB at bottom: 1 = upper-left 3V3; 19 = lower-left 5V; 20 = lower-right CLK; 38 = upper-right GND. Numbers follow Ai-Thinker Table 2; they are board header pins, not module pads or chip pins. Confirm silkscreen on the actual board; other ESP32 boards differ.

Physical / label	GPIO	Direction*	Peripheral / function	Pico target / signal	Wiring / implementation notes
37 / P23	GPIO23	OUT	SPI3 / VSPI MOSI	GP8 / pin 11 SPI1 RX	ESP32 MOSI to Pico RX. Controller-to-peripheral data.
11 / P27	GPIO27	OUT	GPIO / software CS	GP9 / pin 12 SPI1 CSn	Active-low CS; idle HIGH. Recommended: 10 kOhm pull-up to Pico 3V3 to hold CS inactive during ESP32 reset.
30 / P18	GPIO18	OUT	SPI3 / VSPI SCLK	GP10 / pin 14 SPI1 SCK	ESP32 supplies the SPI clock; set mode and rate explicitly to match the Pico protocol.
31 / P19	GPIO19	IN	SPI3 / VSPI MISO	GP11 / pin 15 SPI1 TX	Pico TX to ESP32 MISO. Peripheral-to-controller data.
9 / P25	GPIO25	IN	GPIO / TIME_SYNC	GP12 / pin 16 TIME_SYNC	Qualified, software-delayed indication from Pico of an accepted authoritative UTC/PPS boundary.
10 / P26	GPIO26	TBD / reserve	GPIO / control or IRQ	GP13 / pin 17 Reserved control / IRQ	GPIO reserved only; direction and polarity remain TBD. Leave unwired and undriven until defined.
32 / GND	-	Reference	Shared ground	GND / pin 13 or 18	Connect board grounds. NodeMCU header pins 14 and 38 are also GND. Power distribution is not specified here.

*Directions are relative to the ESP32. ESP32 is the SPI controller; Pico/RP2350 is the native SPI1 peripheral. SPI3 / VSPI names the ESP32 bus; Pico SPI1 is a separate peripheral on the other board.

Protocol v1 / TIME_SYNC: Follow the Pico project's docs/clock-network-protocol.md for qualification and delay reporting. TIME_SYNC is not raw PPS, a zero-delay signal, or a calibrated precision output. Its edge/pulse behavior is governed by that protocol; this sheet adds no timing guarantee.

Electrical / implementation: Shared ground and 3.3 V logic only; no 5 V on signal pins. Keep interconnects short with a nearby ground return. Avoid driving an unpowered board. Set GPIO numbers explicitly in firmware; assert software CS only for a complete transaction. Leave GPIO26 high-impedance with pulls disabled.

Pin selection: GPIO23/19/18 are the SPI3 MOSI/MISO/SCLK IO_MUX pins. Software CS on GPIO27 avoids the default GPIO5 strap. All six selected GPIOs avoid flash GPIO6-11, straps GPIO0/2/5/12/15, and UART0 GPIO1/3. GPIO25/26/27 are used digitally, not as ADC2 inputs. SPI mode, rate, framing and timing must be verified against both firmware implementations.

Interface foundation: Absurdly Accurate Clock - Pico 2 / RP2350 pin-assignment reference, Rev 2, refreshed 26 Sep 2026 (absurdly-accurate-clock-pin-assignment-rev2-refreshed-2026-09-26.pdf). Pico GP8-GP13 assignments are unchanged.

References checked 26 Sep 2026: Ai-Thinker NodeMCU-32S V1 (2019), Fig. 1 / Table 2; Espressif WROOM-32 v3.8, pin definitions / boot configuration; ESP-IDF SPI and GPIO guides. Documentation-verified design; no bench qualification performed.

Ai-Thinker board datasheet (original URL; now redirects): https://docs.ai-thinker.com/_media/esp32/docs/nodemcu-32s_product_specification.pdf

Manufacturer PDF, preserved copy inspected: https://github.com/xiaolaba/ESP32-CAM_blink/blob/master/nodemcu-32s_product_specification.pdf

Espressif ESP32-WROOM-32 datasheet: https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32_datasheet_en.pdf

Espressif ESP-IDF: [SPI Master Driver](#) - [GPIO Matrix and IO_MUX](#) | [GPIO & RTC GPIO](#) - restrictions (clickable source links)